[PROJECT DOCUMENTATION]

CareFlow V1

IIoT Industrial Energy Monitoring System

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COMPANY	Southern IoT, Dhaka	CATEGORY	IIoT / Industrial Systems
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Technology Stack:

RAK3172 LoRaWAN ESP32 Modbus RTU/RS485 3D Printing Machine Learning Dashboard UI

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§1 Project Overview

> This document is the complete archive of all work performed by the Implementation Lead on the CareFlow V1 project at Southern IoT. It covers enclosure design, deployment research, field site visits, machine compatibility testing, and the CareFlow dashboard concept and design.

§1.1 What Is CareFlow V1?

CareFlow V1 is an Industrial IoT (IIoT) energy monitoring system developed by Southern IoT. It is designed to be deployed in industrial plants and commercial facilities to monitor multi-phase electrical parameters from various types of machinery — including generators, boilers, compressors, and UPS systems — in real time.

The system uses **Modbus RTU/RS485** to extract electrical data from existing industrial devices, transmits data over a **LoRaWAN** network using the **RAK3172** module, and aggregates readings on an **ESP32** for local processing and cloud forwarding. The goal is to provide industrial clients with a production-ready, plug-and-play monitoring solution that requires minimal on-site configuration.

§1.2 Implementation Lead's Scope of Work

As Implementation Lead, the following areas were personally owned and executed:

- Enclosure design and 3D printing for field-deployable CareFlow V1 hardware units
- Research and development toward a one-tap deployment workflow
- Field visits to six industrial and commercial client sites to collect real-world data on system behaviour and failure modes
- Hardware compatibility testing across multiple industrial machine types
- Dashboard concept, architecture, and UI design for the CareFlow monitoring platform

§2 Enclosure Design & 3D Printing

§2.1 Design Objective

The enclosure for CareFlow V1 was designed to house all electronics in a compact, protected unit suitable for deployment in industrial environments. The requirements were:

- Fit the ESP32, RAK3172 LoRaWAN module, Modbus RS485 transceiver, and power regulation on a single internal mounting plate
- Provide strain relief and gland entries for RS485 signal cables and power cables
- Allow antenna clearance for the LoRaWAN RAK3172 module (external SMA or PCB antenna)
- Be easy to open for field servicing without special tools
- Carry Southern IoT branding and a professional finished appearance

§2.2 Design & Print Process

The enclosure was designed using 3D CAD software and printed in-house. Key design decisions:

- **Material:** PLA / ABS — selected based on rigidity requirements for wall-mounted industrial deployment
- **Form factor:** Compact rectangular box with snap-fit or screw-lid top panel
- **DIN rail compatibility:** Internal standoffs aligned to standard board mounting holes
- **Cable entry:** Bottom-entry gland holes to prevent dust and water ingress from above
- **LED window:** Frosted light pipe slot on front face for status LED visibility without opening the enclosure

The design went through multiple print iterations to resolve fitment issues, particularly around the PCB standoff spacing and the cable gland sizing. Final dimensions were validated against the assembled board before production printing.

§2.3 3D Model Reference

[INSERT: Enclosure front panel photo]

Fig 1 · CareFlow V1 enclosure — front view

[INSERT: Enclosure interior photo]

Fig 2 · Interior — PCB, RAK3172, RS485 board

[INSERT: 3D CAD model render]

Fig 3 · CAD model — isometric view

[INSERT: Installed unit on site]

Fig 4 · Deployed unit — site installation

GrabCAD / 3D File: grabcad.com · CareFlow V1 Enclosure [ADD LINK]

§3 One-Tap Deployment Research

§3.1 Problem Statement

A key challenge with industrial IoT deployments is the time and expertise required on-site. Every additional step in the commissioning process increases the risk of misconfiguration, extends site visit duration, and demands skilled personnel. The goal was to reduce the CareFlow V1 deployment to a single-action process: **install the unit, connect cables, power on — done.**

§3.2 Research Areas Investigated

The following areas were researched to achieve one-tap deployment:

Auto-Provisioning of LoRaWAN Credentials

Investigated methods to pre-provision LoRaWAN OTAA keys (DevEUI, AppEUI, AppKey) into NVS flash at the factory stage, eliminating the need for on-site key entry via serial or BLE. Studied Espressif's NVS partition pre-flashing workflow and RAK3172 AT command set for pre-configuration.

Modbus Device Auto-Detection

Researched a device scanning routine that sweeps Modbus RTU addresses (1–247) on startup to auto-detect connected slave devices and their register maps. This removes the need for a technician to manually enter Modbus slave address, baud rate, and register offsets during installation.

Wi-Fi / Network Fallback Logic

Developed a resilient network connection strategy:

- Primary path: LoRaWAN for long-range, low-power uplink
- Fallback path: Wi-Fi to local gateway when LoRaWAN link quality degrades
- Offline buffer: Data queued in ESP32 SPIFFS/NVS during network outage, flushed when connectivity restores

Findings from Site Visits

Real-world site visits directly informed this research. The most common installation-blocking issues observed across sites were:

Issue	Implication for Deployment
Power interruption (loadshedding)	Device must recover automatically — no manual restart
RS485 address conflicts	Auto-scan needed to identify device address without prior knowledge
Weak LoRaWAN signal indoors	Signal strength must be assessed at install time; antenna placement matters
No IT personnel on site	All config must be pre-loaded; no on-site terminal access
Varying baud rates across machines	Firmware must auto-detect baud rate (9600 / 19200 / 38400)

§4 Site Visits & Field Experience

§4.1 Overview

Six industrial and commercial sites were visited to test CareFlow V1, gather real-world operational data, and identify deployment challenges specific to each environment. These visits provided direct exposure to the conditions the system must operate reliably within.

§4.2 Sites Visited

Site	Type	Primary Learning
Hamid Tex	Textile manufacturing	High-power 3-phase loads on industrial looms; voltage sag events during motor startup; RS485 interference from VFDs
Crown Cotton	Cotton processing facility	Long cable runs from MCB panels to monitoring point; baud rate inconsistencies across Modbus-capable controllers
Crown Resort	Commercial hospitality	Multiple separate consumer loads (HVAC, lighting, water heating); need for sub-meter tracking per floor/zone
Southern Clothings	Garment manufacturing	Frequent power-cut and generator switchover events; critical to test auto-recovery of LoRaWAN and Modbus after hard power reset
Southern Service	Service / maintenance facility	Mixed AC/DC load environment; tested compatibility with UPS and DC-powered auxiliary systems
The People's Diagnostic Center	Medical facility	Strict uptime requirements; tested uninterrupted monitoring through generator switchover; validated that zero data loss occurs during brief power transitions

§4.3 Key Observations Across All Sites

Power Reset Behaviour

At every site, loadshedding (planned or unplanned power cuts) was encountered. The key learning: the device must complete a full reconnect — LoRaWAN join, MQTT session, and Modbus poll cycle — within 30 seconds of power restoration, without any manual intervention. This directly drove the firmware auto-reconnect logic and the NVS-based session state persistence.

Packet Loss on LoRaWAN

Indoor environments with concrete construction (particularly at Hamid Tex and Crown Cotton) showed significant LoRaWAN signal attenuation. Packet loss rates of 15–30% were measured at certain panel locations. This validated the need for a local data buffer — any reading that fails to transmit is queued and retried, ensuring no gap in the energy log.

Modbus Interference

At Hamid Tex, Variable Frequency Drives (VFDs) on the same RS485 bus introduced noise that corrupted Modbus responses at intervals. This led to implementing CRC validation on every frame and a retry-with-backoff strategy for failed reads.

Generator Switchover at Diagnostic Center

The People's Diagnostic Center provided the most demanding test environment: a medical facility that cannot tolerate monitoring gaps during generator cutover. The switchover takes approximately 3–4 seconds. Testing confirmed that the firmware's NVS write-before-sleep strategy preserved the last known readings across the outage, and the device resumed from correct state on reconnection.

§5 Machine Compatibility Testing

§5.1 Objective

CareFlow V1 must extract data from existing industrial equipment using Modbus RTU over RS485 — without modifying or interrupting the machine's operation. Different machine types expose different register maps, baud rates, and data formats. Compatibility was tested across the following machine categories:

§5.2 Machines Tested

Machine Type	Interface	Parameters Extracted / Notes
Genset Controller	Modbus RTU / RS485	Active power (kW), reactive power (kVAR), apparent power (kVA), voltage L1/L2/L3, current per phase, frequency, runtime hours. Tested on DeepSea and ComAp controller families. Register maps differ significantly — required per-model profile.
Boiler Machine	Modbus RTU / RS485	Fuel consumption rate, operating pressure, temperature setpoint vs actual, run/fault status register. Some boilers use non-standard function codes (FC03 vs FC04) — required firmware to support both read holding and read input registers.
Compressor	Modbus RTU / RS485	Load/unload cycles, discharge pressure, motor current, fault code register, total run hours. Compressor units at Hamid Tex were on address 0x01 with 19200 baud — confirmed auto-baud detection correctly identified the rate.
UPS	Modbus RTU / RS485	Input voltage, output voltage, battery charge %, load percentage, bypass status. UPS units at Southern Service and Diagnostic Center both tested successfully. Key finding: some UPS Modbus implementations use 16-bit scaled integers (÷10) for voltage — the firmware scaling layer handles this.

§5.3 Testing Outcomes

- All four machine types successfully polled and data extracted without interrupting normal machine operation
- Modbus function codes FC01, FC02, FC03, FC04 all implemented and verified across device types
- Baud rate auto-detection confirmed working at 9600, 19200, and 38400 bps
- Register map profiling system designed: each machine type gets a JSON profile defining address, scaling, and unit — loaded from flash at boot
- Firmware CRC error rate on live machines: $< 2\%$ after noise-mitigation improvements

§6 CareFlow Dashboard — Concept & Design

§6.1 Design Objective

The CareFlow dashboard was conceived and designed to present industrial energy data in a way that is **actionable** at three distinct user levels. The core design principle is that each role sees exactly what it needs — no more, no less — to make effective decisions within its scope of responsibility.

§6.2 Three-Tier Access Architecture

The dashboard implements a role-based access control (RBAC) model with three tiers:

[ROLE 1] **Admin — Plant Overview**

The Admin has full visibility across the entire facility. The dashboard for this role presents:

- **Plant-wide energy map:** All monitored machines displayed in a single overview panel with real-time kW, kVAR, and kWh per device
- **Consumption trend charts:** Daily, weekly, and monthly energy graphs per machine and per facility total
- **Demand peak alerts:** Visual flag when any machine or the total plant demand exceeds a configurable threshold
- **Device health overview:** Online/offline status of all CareFlow nodes at a glance
- **Export & reporting:** CSV/PDF energy report generation for billing and compliance
- **User management:** Add/remove Technician and Worker accounts; assign them to specific machines or zones

[ROLE 2] **Technician — ML-Driven Alerts**

The Technician is a trained industrial professional responsible for investigating anomalies. The system applies a **machine learning model** to the continuous sensor stream and flags deviations from learned baseline behaviour. The Technician dashboard presents:

- **Anomaly alert feed:** Live list of ML-flagged events (e.g. abnormal current draw, unusual power factor deviation, unexpected runtime patterns)
- **Alert detail view:** Per-alert waveform overlay comparing current reading against learned baseline
- **Machine drill-down:** Full register dump and trend chart for any single device
- **Alert triage:** Confirm (escalate to Worker) or dismiss (log as false positive for model retraining) each alert

- **Maintenance log:** Record of all confirmed and resolved faults with timestamps

The ML layer uses **anomaly detection** trained on historical baseline readings per machine — flagging statistically significant deviations without requiring threshold pre-configuration for every parameter.

[ROLE 3] **Worker — Confirmed Alert Notifications**

The Worker is the on-floor operator who acts on machine alerts. They do not interpret raw data — they receive only **Technician-confirmed alerts** that require a physical response. The Worker interface is deliberately simplified:

- **Alert notification:** Push notification (mobile) or audible/visual signal on the floor indicating which machine requires attention
- **Action card:** Plain-language description of the confirmed issue and recommended action (e.g. "Compressor #3 — abnormal current. Check belt tension.")
- **Acknowledgement:** Worker marks the task as attended; this is logged and visible to the Technician and Admin
- **No raw data exposure:** Raw registers, charts, and configuration are all hidden from the Worker role — reducing cognitive load and preventing accidental changes

§6.3 Dashboard Visual Design Concepts

[INSERT: Admin dashboard mockup/screenshot]

Fig 5 · Admin dashboard — plant overview panel

[INSERT: Technician alert feed mockup]

Fig 6 · Technician dashboard — ML alert feed

[INSERT: Worker notification view mockup]

Fig 7 · Worker view — confirmed alert card

[INSERT: Energy trend chart / graph]

Fig 8 · Admin trend view — daily kWh per machine

§6.4 ML Integration Architecture (Planned)

The machine learning component is planned to operate as follows:

1. **Baseline learning phase (Days 1–14):** The system collects readings from each machine over a rolling 2-week window and builds a statistical baseline (mean, standard deviation, and time-of-day profile per parameter)
2. **Anomaly scoring:** Each new reading is scored against the baseline using a Z-score or Isolation Forest model. Scores above a configurable threshold trigger a Technician alert
3. **Feedback loop:** Technician dismissals (false positives) are fed back to the model as negative examples, improving precision over time
4. **Model hosting:** Lightweight model inference runs server-side (cloud or local edge server); ESP32 only transmits raw readings — no on-device ML required

§7 Current Status Summary

Item	Status
Enclosure design (3D CAD) & print	Complete
Site visits — 6 sites completed	Complete
Machine compatibility testing (4 types)	Complete
One-tap deployment research	Research complete
Firmware auto-reconnect after power cut	Implemented
Modbus multi-device polling (FC01–FC04)	Implemented
LoRaWAN offline buffer & retry	Implemented
Dashboard architecture & role design	Concept complete
Dashboard UI mockups (Admin/Tech/Worker)	In progress
ML anomaly detection model	In research
Dashboard backend development	Planned
CareFlow V2 hardware revision	Planned

Remarks: CareFlow V1 has been successfully deployed and validated across six industrial and commercial sites. All critical field issues — power reset recovery, Modbus interference, LoRaWAN packet loss, and machine compatibility — have been identified, documented, and resolved in firmware. The dashboard concept is complete with a clear three-tier access model; development and ML integration are the next milestone. The enclosure is production-ready and has been validated across all visited sites.

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